



FINCHAIN: A Symbolic Benchmark for Verifiable Chain-of-Thought Financial Reasoning

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Abstract

Multi-step symbolic reasoning is essential for robust financial analysis; yet, current benchmarks largely overlook this capability. Existing datasets such as FinQA and ConvFinQA emphasize final numerical answers while neglecting the intermediate reasoning steps required for transparency and verification. To address this gap, we introduce FINCHAIN, the first benchmark specifically designed for verifiable Chain-of-Thought evaluation in finance. FINCHAIN spans 58 topics across 12 financial domains, each represented by parameterized symbolic templates with executable Python code that enable fully machine-verifiable reasoning and scalable, contamination-free data generation. To assess reasoning capacity, we propose CHAINEVAL, a dynamic alignment measure that jointly evaluates both the final-answer correctness and the step-level reasoning consistency. Our evaluation of 26 leading LLMs reveals that even frontier LLMs exhibit clear limitations in symbolic financial reasoning, while domain-adapted and math-enhanced fine-tuned models can substantially narrow this gap. Overall, FINCHAIN exposes persistent weaknesses in multi-step financial reasoning and provides a foundation for developing trustworthy, interpretable, and verifiable financial AI. This project is available at <https://github.com/mbzuai-nlp/finchain.git>.

1 Introduction

Large language models (LLMs) have demonstrated strong performance across a wide range of tasks (Zhao et al., 2023; Xie et al., 2023b).

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FINCHAIN (Compound Interest)

#Question:

`investor_name` invested `principal` in `project_name`. The investment grows at an annual interest rate of `rate`% compounded annually over `time` years. Calculate the compound interest (CI).

#Variables:

- `investor_name` = sample(investors)
- `project_name` = sample(projects)
- `principal` = range(1000, 5000)
- `rate` = uniform(2, 10)
- `time` = range(1, 5)

#Chain-of-Thought Solution:

Step 1: Compute the compound amount: `amount` = `principal` ×

$$\left(1 + \frac{\text{rate}}{100}\right)^{\text{time}}$$

Step 2: Compute the compound interest:

$$\text{CI} = \text{amount} - \text{P}$$

Figure 1: Symbolic template for generating compound interest problems in FINCHAIN.

These models have likewise shown promise in financial applications (Chen et al., 2024b; Xie et al., 2026), where effective analysis often requires synthesizing large volumes of textual information from reports, news, and social media, which reflect and influence financial phenomena such as investor sentiment, risk perceptions, and expected market trends (Nie et al., 2024; Zhang et al., 2026; Zhou et al., 2026).

Most prior work in financial NLP has focused on tasks with shallow supervision, including information extraction (Shah et al., 2023), sentiment analysis (Pei et al., 2022), and text classification (Sy et al., 2023). These tasks typically require models to produce short outputs and do not test whether they can perform transparent, multi-step financial reasoning. In contrast, many financial problems require generating structured chains of reasoning that justify each intermediate step, as illustrated in Figure 1. Existing financial reasoning benchmarks such as FinQA (Chen et al., 2021) and ConvFinQA (Chen et al., 2022) primarily frame reasoning as numerical question answering and emphasize final-answer correctness. While some examples include intermediate reasoning traces, these are neither systematically structured nor rigorously verified. As a result, such benchmarks cannot reliably diagnose where reasoning fails or distinguish genuine multi-step inference from surface-level pattern matching.

Inspired by the symbolic-template paradigm introduced in mathematical reasoning (Mirzadeh et al., 2025), we construct a financial reasoning benchmark entirely from scratch. As shown in Figure 1, each symbolic template encodes a parameterized financial problem with named variables and numeric inputs, paired with executable Python code that computes both intermediate steps and final results. This design supports scalable, contamination-free data generation grounded in explicit symbolic and numerical operations. Financial reasoning spans diverse topics and requires heterogeneous expertise. To capture this diversity, we organize our dataset using a fine-grained taxonomy (see Figure 2) covering 12 domains and 58 topics. For each topic, we design five parameterized templates of increasing difficulty, comprising two easy, two intermediate, and one advanced template. Each instantiated example consists of a scenario card specifying the topic, the difficulty, and some example inputs, together with an executable chain of reasoning steps grounded in domain-specific formulae. Because the gold reasoning traces are explicit and executable, the intermediate computations and the final results can be verified at the symbolic and the numerical levels, thus enabling automatic detection of incorrect or inconsistent reasoning steps. To support rigorous and interpretable evaluation, we introduce CHAINEVAL, a dynamic-alignment metric that jointly evaluates final-answer correctness and intermediate step faithfulness.

Unlike conventional text similarity measures, CHAINEVAL explicitly accounts for both semantic correspondence and numerical consistency between predicted and reference reasoning chains. Using this benchmark and evaluation framework, we evaluate 26 proprietary and open-weight LLMs. We find that frontier LLMs perform best overall, yet consistently struggle with advanced multi-step symbolic financial reasoning, while fine-tuned compact models achieve only limited gains.

Our main contributions are as follows:

- We introduce the first from-scratch symbolic benchmark for financial reasoning, grounded in a fine-grained taxonomy spanning 12 domains and 58 topics.
- We propose CHAINEVAL, a verifiable reasoning measure that evaluates both step-level consistency and final-answer correctness, and shows the strongest correlation with expert human judgments.
- We benchmark 26 leading proprietary and open-weight LLMs, and find that even state-of-the-art LLMs struggle with verifiable multi-step financial reasoning, particularly on advanced symbolic templates.

2 Related Work

2.1 Financial NLP

Progress in financial NLP has been driven by both modeling and benchmarking. Early work focused on extraction and classification with models such as FinBERT (Liu et al., 2020), while later efforts expanded to personal finance (Hean et al., 2025), credit scoring (Feng et al., 2023), and risk-awareness benchmarking (Yuan et al., 2024). Datasets like FiNER-ORD, REFinD, FinARG, and ECTSum support tasks in NER, relation extraction, argument mining, and summarization (Shah et al., 2023; Kaur et al., 2023; Mukherjee et al., 2022; Xie et al., 2024). Large financial language models have further advanced the field. BloombergGPT (Wu et al., 2023) achieved broad in-domain performance, FinGPT (Liu et al., 2023) emphasized open-source adaptability, and FinMA (Xie et al., 2023a) delivered competitive results with a compact architecture. Corresponding benchmarks such as FLANG (Shah et al., 2022), FinBen (Xie et al., 2024), and FinMTEB (Tang and Yang, 2025) broadened evaluation coverage across diverse tasks.

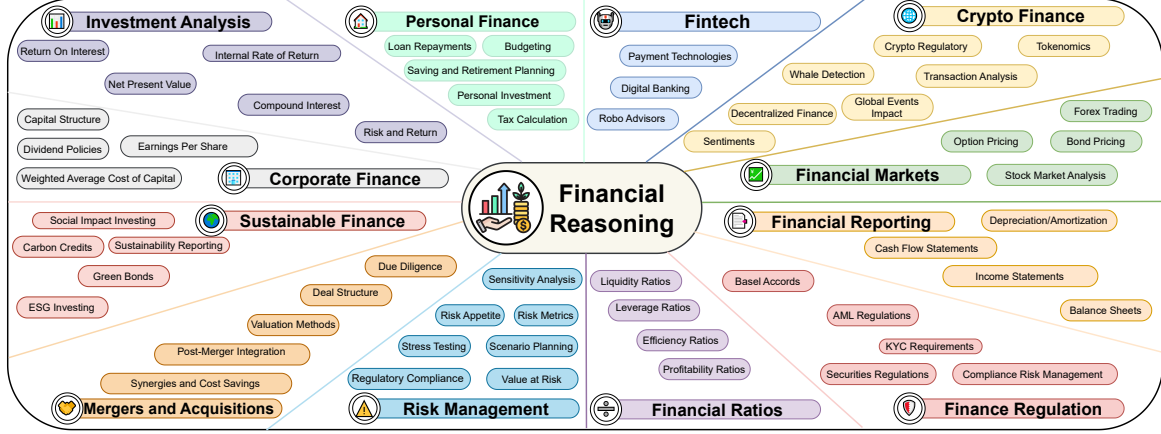


Figure 2: **FINCHAIN taxonomy of financial reasoning topics.** Our benchmark spans 58 topics organized into 12 major domains, ranging from traditional areas like *Corporate Finance* and *Financial Reporting* to emerging fields such as *Crypto Finance* and *Sustainable Finance*. This hierarchical structure enables fine-grained evaluation of symbolic reasoning across diverse financial domains.

Furthermore, BizBench (Krumdick et al., 2024) and PIXIU (Xie et al., 2023a) evaluated LLMs on quantitative and multimodal reasoning. Despite this progress, limitations remain in multi-step reasoning, long-context understanding, and cross-market generalization (Chen et al., 2024b). These gaps motivate the need for benchmarks that assess the capability of LLMs to perform faithful, auditable reasoning grounded in financial knowledge.

2.2 Financial Reasoning

Real-world financial problems require precise numerical reasoning. FinQA (Chen et al., 2021) and ConvFinQA (Chen et al., 2022) were developed before Chain-of-Thought reasoning became a standard evaluation target. As a result, they supervise arithmetic program generation but offer only weak step-level signals, yielding traces that are neither explicit nor verifiable. FinTextQA (Chen et al., 2024a) introduces long-form financial questions from textbooks and regulatory sources and focuses on explanatory retrieval rather than traceable computation. Bridging text and numerical reasoning, TAT-QA (Zhu et al., 2021) and MultiHiertt (Zhao et al., 2022) combine textual and tabular evidence, while DocMath-Eval (Zhao et al., 2024b) and FinanceMath (Zhao et al., 2024a) move toward interpretable symbolic evaluation. However, these datasets remain largely domain-agnostic and lack explicit, step-level supervision grounded in financial formulae. More recently, FinanceReasoning (Tang et al., 2025) improves answer-level numerical reliability by introducing executable Python solutions.

However, FinanceReasoning does not provide systematic verification of step-level reasoning alignment. FINCHAIN addresses this gap by introducing a symbolic, executable benchmark with explicit intermediate supervision and automatic alignment-based evaluation, spanning 58 topics across 12 financial domains.

3 FINCHAIN

3.1 Data Creation Process

We begin by identifying and defining financial domains based on established literature (Bodie et al., 2025) and expert input within the team, resulting in 12 distinct domains. Within each domain, we propose candidate financial topics with LLM assistance and curate them with financial experts, yielding a total of 58 topics (mean 4.8 per domain). The resulting taxonomy is illustrated in Figure 2. Following Mirzadeh et al. (2025), we instantiate each topic through parameterized symbolic templates that define both the question structure and an executable Chain-of-Thought solution grounded in domain-specific formulas. We implement these templates as executable Python programs that generate both intermediate reasoning steps and final answers, thereby enabling fully machine-verifiable evaluation. For each topic, we design five templates spanning three difficulty levels (two basic, two intermediate, and one advanced), where we control difficulty by the number and the complexity of required reasoning steps. Table 1 summarizes the dataset statistics and shows increasing reasoning depth across difficulty levels.

Statistic	Basic	Intermediate	Advanced
#Templates	116	116	58
Avg. steps	2.01	2.97	3.90

Table 1: Dataset statistics of FINCHAIN.

An example symbolic template is shown in Figure 1. We generate the templates with LLM assistance and subsequently curate them with domain experts to ensure correctness, consistency, and balanced difficulty. We provide detailed prompt designs and generation procedures in Appendix A.1. This design isolates financial reasoning ability from document parsing challenges. We therefore position FINCHAIN as a controlled testbed for verifiable financial reasoning, complementary to benchmarks centered on real-world document understanding.

3.2 Data Validation and Expert Review

To ensure data quality and consistency, we apply a set of validation constraints covering numerical precision, unit consistency, input completeness, and reasoning step informativeness. Templates that fail validation are revised prior to expert review. A detailed description of the validation criteria is provided in Appendix A.3. We ask financial experts to review all validated templates following a calibrated annotation protocol. Specifically, all reviewers first participated in a pilot calibration phase, during which they jointly reviewed a shared subset of templates and discussed discrepancies to align on annotation standards. After this calibration phase, reviewers independently assessed the remaining templates, evaluating both the correctness of reasoning steps and the final numerical results under the agreed-upon criteria. Further details on annotator backgrounds, annotation procedures, and quality control are provided in Appendix A.4.

4 CHAINEVAL

We propose CHAINEVAL, an evaluation framework that jointly assesses reasoning-step alignment and final-answer correctness. Building on prior work on reasoning consistency (Lyu et al., 2023; Golovneva et al., 2023), our approach explicitly verifies intermediate results via step-answer matching while also checking the final numerical outcome.

4.1 Preliminaries

We define the gold solution S^* and the predicted solution $\hat{S}$ as sequences of m and n reasoning steps, respectively:

$$S^* = (s_1^*, \dots, s_m^*), \quad \hat{S} = (\hat{s}_1, \dots, \hat{s}_n), \quad (1)$$

where s_i^* and $\hat{s}_j$ denote individual reasoning steps in the gold and in the predicted solutions, respectively. Each step s_i produces an intermediate result,

$$\text{StepRes}(s_i) = a_i, \quad (2)$$

representing the numerical or the symbolic value computed at that step.

To evaluate the reasoning faithfulness, we compare these sequences both semantically and numerically. In addition, we apply *Dynamic Time Warping* (DTW) to capture the global structural alignment between step sequences. DTW provides an order-preserving but flexible alignment that accommodates insertions, deletions, or small reordering of steps while maintaining the overall sequence coherence.

4.2 Reasoning Step Alignment

We assess the consistency between gold and predicted reasoning traces through two complementary components: semantic similarity and answer-level agreement, combined within a DTW-based alignment framework.

Semantic Similarity. Each step is encoded using a sentence encoder $\text{Enc}(\cdot)$, and the pairwise semantic similarity between the gold and the predicted steps is computed as

$$\text{SS}(s_i^*, \hat{s}_j) = \cos(\text{Enc}(s_i^*), \text{Enc}(\hat{s}_j)), \quad (3)$$

where $\cos(\cdot, \cdot)$ denotes the cosine similarity and $\text{SS} \in [0, 1]$.

Answer Match. For the intermediate results produced by each step, we evaluate numeric or symbolic consistency:

$$\text{StepRes}(s_i^*) = a_i^*, \quad \text{StepRes}(\hat{s}_j) = \hat{a}_j.$$

We then define the answer-matching function:

$$\text{AM}(s_i^*, \hat{s}_j) = \begin{cases} \mathbb{I}\left(\frac{|\hat{a}_j - a_i^*|}{|a_i^*|} \leq \epsilon\right), & \text{if both are numeric,} \\ \mathbb{I}(\hat{a}_j = a_i^*), & \text{otherwise.} \end{cases} \quad (4)$$

Here, $\mathbb{I}(\cdot)$ denotes the indicator function, and $\epsilon = 0.05$ permits up to a 5% relative numerical deviation to account for rounding or error propagation. This design choice is motivated by financial auditing standards¹, in which materiality thresholds are commonly defined as between 5% and 10% of a base metric such as earnings, and deviations below 5% are generally considered immaterial.

Gated Step-Level Similarity. To ensure that a pair of steps is considered consistent only when both their semantics and results agree, we define a gated score:

$$\text{Score}_{\text{gate}}(i, j) = \text{SS}(s_i^*, \hat{s}_j) \times \text{AM}(s_i^*, \hat{s}_j). \quad (5)$$

This score forms the basis of the DTW alignment matrix.

Dynamic Sequence Alignment. To capture global reasoning consistency, we align the two step sequences using *Dynamic Time Warping* (DTW). DTW searches for an optimal monotonic path between $(S^*, \hat{S})$ that minimizes cumulative cost while preserving step order. This formulation allows local insertions, deletions, and compressions, and does not require strict template matching as long as the predicted reasoning remains semantically and numerically aligned with the reference.

DTWGate Alignment Score. We transform the minimal DTW cost into a normalized similarity measure as follows:

$$\text{DTWGate}(S^*, \hat{S}) = 1 - \frac{\text{Cost}_{\text{DTW}}}{L_{\text{path}}}, \quad (6)$$

where Cost_{DTW} denotes the total alignment cost and L_{path} represents the length of the optimal alignment path. The resulting score lies in the range $[0, 1]$, with higher values indicating stronger reasoning alignment between the gold and the predicted solutions.

4.3 Final Answer Correctness

Beyond step-level reasoning alignment, we also assess the correctness of the final predicted outcome. Let s_m^* and $\hat{s}_n$ denote the last steps of the gold and the predicted solutions, respectively. We define the **Final Answer Correctness (FAC)** metric as

$$\text{FAC}(S^*, \hat{S}) = \begin{cases} \mathbb{I}\left(\frac{|\hat{a}_n - a_m^*|}{|a_m^*|} \leq \epsilon\right), & \text{if both are numeric,} \\ \mathbb{I}(\hat{a}_n = a_m^*), & \text{otherwise,} \end{cases} \quad (7)$$

¹<https://www.materialitytracker.net/standards/financial-thresholds/>

Here, we use the same tolerance $\epsilon = 0.05$ as before. FAC measures whether the model’s final computation aligns with the correct end result, complementing the DTW-based metric that evaluates reasoning faithfulness throughout the entire solution sequence. Therefore, we have

$$\text{CHAINEDVAL} = (1 - \alpha) \cdot \text{DTWGate} + \alpha \cdot \text{FAC}, \quad (8)$$

which accounts for both reasoning correctness and final answer correctness. We set $\alpha = 0.1$, selected via grid search on a subset to maximize the correlation with human evaluations, as explained in [Appendix D](#). We further verify that the final measure best reflects true reasoning quality by comparing it with human evaluations, as described in § 5.2.

5 Experiments and Results

5.1 Evaluated Models

We evaluate a total of 26 LLMs, grouped into four categories according to their capability and relevance to financial reasoning. (1) **Frontier proprietary models**, including GPT-5, 4.1, 5-mini, 4.1-mini ([OpenAI, 2025a,b](#)), Claude Sonnet {4.5, 4, 3.7} ([Anthropic, 2025b,c,a](#)), Gemini-2.5 {Pro, Flash} ([Comanici et al., 2025](#)), DeepSeek-{V3.2, V3.1, R1} ([Liu et al., 2024](#); [Guo et al., 2025](#)), and Grok-4 {Heavy, Fast} ([xAI, 2025](#)). (2) **Finance-specific models**, including Fin-o1 ([Qian et al., 2025](#)), Fin-R1 ([Liu et al., 2025](#)), DianJin-R1 ([Zhu et al., 2025](#)), and WiroAI Finance-{LLaMA, Qwen} ([Bezir et al., 2025](#)). (3) **Math-enhanced models**, including WizardMath ([Luo et al., 2025](#)), MetaMath ([Yu et al., 2023](#)), Mathstral ([Mistral AI, 2024](#)), and Qwen-2.5-Math ([Yang et al., 2024](#)). (4) **General-purpose open-weight models**, including LLaMA-3.1 ([Grattafiori et al., 2024](#)) and Qwen-{2.5, 3} ([Qwen, 2024](#); [Qwen Team, 2025](#)). Detailed configurations and model sources are described in [Appendix C](#).

5.2 CHAINEDVAL Validation

Before conducting large-scale experiments, we validated the proposed CHAINEDVAL metric through a controlled expert evaluation. We randomly sampled 20 instantiated questions from the FINCHAIN dataset and generated answers using five models of different capacities and training paradigms, namely GPT-5, GPT-4.1 mini, MetaMath, Fin-o1, and LLaMA-3.1, to ensure a clear range of reasoning quality, producing 100 model-generated responses.

Model	Size	CHAI _{NEVAL} ↑	FAC ↑	ROUGE R ₂ ↑	ROUGE R _L ↑	BERTScore ↑
Frontier Proprietary LLMs						
GPT-5	N/A	66.57 ^{10.64}	82.03 ^{32.40}	28.84 ^{12.30}	42.77 ^{12.91}	88.77 ^{2.39}
GPT-4.1	N/A	65.34 ^{9.36}	84.66 ^{29.26}	19.38 ^{8.91}	30.12 ^{10.57}	86.04 ^{2.09}
GPT-5-mini	N/A	67.17 ^{11.06}	80.28 ^{32.34}	26.48 ^{11.99}	39.74 ^{12.83}	88.18 ^{2.35}
GPT-4.1-mini	N/A	65.06 ^{8.88}	84.59 ^{30.23}	18.67 ^{8.86}	29.05 ^{10.51}	86.05 ^{2.09}
Claude Sonnet 4.5	N/A	66.33 ^{9.44}	83.34 ^{31.79}	19.69 ^{7.77}	29.37 ^{8.81}	86.07 ^{2.00}
Claude Sonnet 4	N/A	66.20 ^{9.66}	82.62 ^{31.96}	19.87 ^{7.82}	29.50 ^{8.59}	86.38 ^{1.83}
Claude Sonnet 3.7	N/A	65.51 ^{9.30}	83.14 ^{31.00}	19.49 ^{7.77}	29.36 ^{8.56}	86.38 ^{1.82}
Gemini-2.5 Pro	N/A	66.04 ^{9.71}	84.34 ^{30.99}	17.61 ^{7.07}	27.36 ^{8.28}	85.94 ^{1.88}
Gemini-2.5 Flash	N/A	65.96 ^{10.10}	83.90 ^{30.61}	18.98 ^{8.05}	29.22 ^{9.32}	86.34 ^{2.02}
DeepSeek-v3.2	N/A	65.23 ^{9.63}	84.17 ^{31.23}	21.73 ^{10.32}	32.85 ^{11.31}	86.66 ^{2.15}
DeepSeek-v3.1	N/A	65.29 ^{9.62}	84.34 ^{31.02}	21.72 ^{10.29}	32.87 ^{11.24}	86.68 ^{2.14}
DeepSeek-R1	N/A	51.22 ^{11.04}	28.97 ^{30.29}	8.67 ^{7.21}	12.93 ^{9.72}	84.39 ^{1.79}
Grok-4 Fast	N/A	60.69 ^{16.26}	66.54 ^{42.54}	21.33 ^{11.09}	32.25 ^{13.30}	86.83 ^{3.06}
Finance Specific LLMs						
Fin-o1	8B	41.50 ^{12.49}	52.79 ^{27.89}	3.47 ^{1.55}	6.35 ^{2.32}	83.55 ^{1.50}
Fin-R1	7B	58.14 ^{7.32}	52.76 ^{28.04}	5.70 ^{2.44}	9.22 ^{3.33}	84.30 ^{1.34}
DianJin-R1	7B	51.95 ^{8.98}	37.69 ^{23.73}	6.28 ^{2.95}	10.79 ^{4.19}	83.12 ^{1.32}
Finance-LLaMA	8B	41.35 ^{10.49}	25.21 ^{25.64}	9.39 ^{4.69}	16.19 ^{5.84}	83.48 ^{2.09}
Finance-Qwen	7B	34.57 ^{11.01}	31.62 ^{25.62}	9.50 ^{4.26}	16.46 ^{5.44}	83.35 ^{1.70}
Math Enhanced LLMs						
WizardMath	7B	24.33 ^{15.00}	41.28 ^{35.69}	11.66 ^{6.57}	20.72 ^{7.83}	84.78 ^{2.36}
MetaMath	7B	7.93 ^{9.43}	23.97 ^{28.76}	11.45 ^{7.36}	21.08 ^{9.24}	84.86 ^{2.99}
Mathstral	7B	59.87 ^{10.02}	54.03 ^{36.61}	16.79 ^{7.82}	26.97 ^{9.34}	86.13 ^{2.18}
Qwen-2.5-Math	7B	55.35 ^{14.98}	62.62 ^{34.56}	11.74 ^{5.87}	20.56 ^{7.61}	83.45 ^{1.85}
General Purpose Open LLMs						
LLaMA-3.1 Instruct	8B	53.99 ^{6.02}	32.72 ^{27.46}	4.61 ^{2.28}	8.09 ^{3.02}	83.35 ^{1.36}
Qwen-2.5 Instruct	7B	60.35 ^{7.47}	65.41 ^{32.53}	9.20 ^{4.51}	15.26 ^{5.85}	84.22 ^{1.78}
Qwen-3	8B	43.32 ^{11.81}	32.28 ^{28.58}	4.05 ^{1.69}	6.61 ^{2.14}	83.58 ^{1.24}

Table 2: **Zero-shot performance across financial, mathematical, and general reasoning benchmarks.** Scores are reported as percentages, with standard deviation in superscript. Model size (N/A) denotes proprietary or undisclosed configurations. Within each model group, the best-performing system for each metric is highlighted in bold.

These responses were then randomly shuffled and anonymized for human assessment. Financial experts independently evaluated each response with respect to the corresponding question and gold-standard reasoning trace. They rated each output along two dimensions, *Reasoning Process Quality* and *Final Answer Accuracy*, on a five-point scale as described in Appendix B.4. We observed a strong association between these dimensions, with Spearman’s ρ exceeding 0.94, showing that coherent reasoning often leads to accurate final answers. Consequently, subsequent analysis below will focus on *Reasoning Process Quality* as the primary evaluation dimension. We further computed several variants of CHAI_{NEVAL} and compared their correlations with expert process scores against reference-based evaluation measures such as ROUGE-2 and ROUGE-L (Lin, 2004). Among all variants, the DTWNormGate formulation with inclusion of final answer’s correctness showed the highest correlation with human judgments, capturing both semantic and numeric consistency across reasoning steps.

Therefore, we adopted it as the primary evaluation measure in this study, as detailed ablation and sensitivity analyses in Appendix D show that it achieves the strongest correlation with human judgments and remains stable across a broad range of ϵ values, with the best result at $\epsilon = 0.05$.

5.3 Experimental Setup

We instantiate the FINCHAIN benchmark by sampling 10 instances per symbolic template with distinct random seeds, yielding $58 \text{ topics} \times 5 \text{ templates} \times 10 \text{ instances} = 2,900$ test cases. We evaluate all models under a unified decoding configuration: temperature = 0.7, top- p = 0.95, and a maximum token limit of 4,096 unless these parameters are unavailable for a given model. We use a zero-shot setup with a standardized reasoning prompt:

Please answer the given question and provide a step-by-step solution.
Use the format: Step 1: ..., Step 2: ..., ...
The question is: {q}

We use CHAINEVAL as the primary evaluation measure, as it jointly measures final-answer correctness and alignment of the intermediate reasoning steps. We post-processed model outputs with regular expressions to extract the ordered list of reasoning steps, accommodating common variations such as “Step x:”, “Step x”, or “stepx”. For comparison, we also report results using frequently used reference-based measures (Xie et al., 2023c) to assess surface-level quality, including ROUGE-2 and BERTScore (Zhang et al., 2020).

5.4 Results

5.4.1 Overall Model Performance

As shown in Table 2, frontier proprietary models achieve the strongest overall performance under CHAINEVAL, indicating higher alignment quality in step-level symbolic reasoning. The performance differences between frontier models are not strictly monotonic. For example, GPT-5-mini slightly outperforms GPT-5, despite comparable or lower scores on surface-level metrics. Among open-source systems, Fin-R1 and Mathstral achieve CHAINEVAL scores of 58.14 and 59.87 respectively, approaching frontier-level alignment despite their substantially smaller scale. These results suggest that targeted fine-tuning and symbolic supervision can substantially improve multi-step reasoning fidelity beyond what model size alone provides. However, the effectiveness of fine-tuning varies with adaptation scope: models trained on broad mathematical corpora (e.g., Mathstral) exhibit stronger generalization across reasoning styles, whereas models tuned narrowly for finance (e.g., Finance-LLaMA and Finance-Qwen) display more variable alignment quality. Notably, the gap between FAC and CHAINEVAL for some reasoning-oriented models suggests that correct final answers do not necessarily imply faithful intermediate reasoning. We also evaluate Grok-4 Heavy, which was among the strongest available systems at the time of evaluation. Due to its high inference cost (approximately \$0.5 per sample), we only assessed this system on a randomly sampled subset rather than the full benchmark, with results reported in Appendix E.1. Overall, these results show that while frontier models lead in verifiable symbolic reasoning, model scale alone is insufficient, and structured supervision plays a critical role in achieving high step-level alignment.

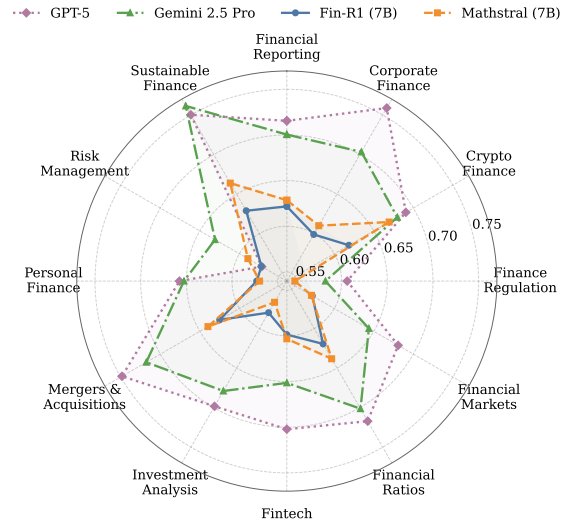


Figure 3: **Domain-level performance across financial domains.** Radar plot showing CHAINEVAL scores across twelve financial domains for four representative models: GPT-5, Gemini-2.5 Pro, Fin-R1, and Mathstral.

5.4.2 Performance Across Domains

Figure 3 shows the domain-level performance for four representative models: GPT-5, Gemini-2.5 Pro, Fin-R1, and Mathstral. Among them, GPT-5 demonstrates consistently strong performance across most domains, forming a relatively stable upper envelope with limited variation. Gemini-2.5 Pro follows a similar trend, achieving competitive scores across domains while exhibiting moderate domain-level differences. The two open-weight models show more heterogeneous performance patterns. Fin-R1 achieves relatively higher scores in domains such as Financial Reporting, Sustainable Finance, and Risk Management, while exhibiting lower performance in several quantitatively intensive or structurally complex domains. In contrast, Mathstral performs competitively in quantitatively orientated domains, such as Financial Ratios and Investment Analysis, but shows weaker performance in domains with heavier textual or regulatory components. In general, domain-level performance is not uniform across models and relative rankings vary by financial category. These observations highlight systematic variation across domains and motivate further diagnostic analysis in subsequent sections. Domain-level results for the remaining models are provided in Appendix E.2. Overall, these results indicate that symbolic financial reasoning ability is highly domain-dependent, and that strong aggregate performance does not guarantee robust reasoning across diverse financial categories.

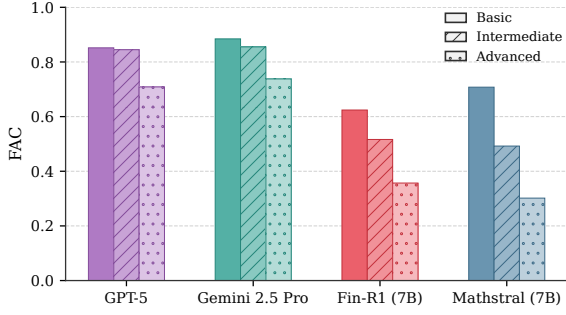


Figure 4: **Final answer correctness (FAC) across difficulty levels.** FAC for representative models on *Basic*, *Intermediate*, and *Advanced* FINCHAIN instances.

5.4.3 Performance Across Difficulty Levels

To assess the model robustness under increasing reasoning complexity, we group FINCHAIN instances into three predefined difficulty tiers: *Basic*, *Intermediate*, and *Advanced* (§ 3). Each tier corresponds to an increase in the number of required reasoning steps and the depth of symbolic and numerical operations. Figure 4 reports model performance across difficulty levels using final answer correctness (FAC), isolating end-task success from partial step-level alignment captured by CHAINEVAL. Across all tiers, frontier proprietary models achieve the highest correctness, with GPT-5 and Gemini-2.5 Pro maintaining relatively strong performance as task difficulty increases. Nevertheless, even these models exhibit a clear degradation on advanced instances, highlighting the challenge of solving complex, multi-step financial reasoning problems to completion. In contrast, open-weight and fine-tuned models show substantially steeper declines as difficulty increases. Mathstral performs competitively on *Basic* and *Intermediate* tasks, suggesting that mathematical fine-tuning improves structured numerical reasoning, but its performance drops markedly on *Advanced* instances. Fin-R1 displays similar behavior, achieving reasonable accuracy on simpler finance-oriented queries while degrading more sharply on tasks requiring longer reasoning chains. Overall, FAC reveals a pronounced gap between frontier and non-frontier models on advanced symbolic reasoning tasks. Taken together with the more gradual degradation observed under CHAINEVAL, these results suggest that completing long, multi-step reasoning chains remains a key bottleneck. We report the corresponding difficulty breakdown under CHAINEVAL in Appendix E.3.

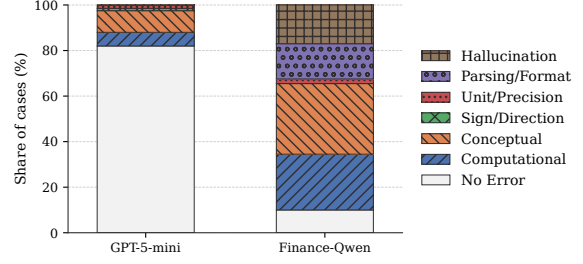


Figure 5: **Error type distribution on a randomly sampled subset of 200 questions.** The figure shows the distributions for GPT-5-mini and Finance-Qwen; the expanded comparison is provided in Appendix F.3.

5.5 Error Analysis

We conduct a targeted error analysis on a randomly sampled subset of 200 questions from FINCHAIN. In Figure 5, we analyze the same question set for GPT-5-mini and Finance-Qwen, which serve as contrasting points on the performance spectrum under an identical evaluation setup, with GPT-5-mini as a strong frontier-aligned reference and Finance-Qwen as a finance-tuned model with lower overall accuracy. This shared subset enables a direct comparison of error profiles without confounding effects from domain coverage or evaluation scale. We further extend the same analysis to six models in Appendix F.3, where we observe the same qualitative patterns across the expanded comparison. We manually inspect model outputs and assign them to a small set of coarse-grained error categories capturing common reasoning failures in financial problem solving. Our taxonomy includes *Computational* errors (incorrect numerical execution), *Conceptual* errors (incorrect application of financial concepts), *Sign/Direction* errors, *Unit/Precision* mismatches, *Parsing/Format* issues that prevent reliable step alignment, and *Hallucination*, which includes content unsupported by the input. Outputs that are fully correct within tolerance are labeled as *No Error*. A detailed description of each category is provided in Appendix F.1. During expert review, a small number of borderline cases initially categorized as computational errors were revised to *No Error* after manual inspection, primarily when numerical deviations were deemed immaterial. All reported statistics reflect these expert-validated labels. Figure 5 summarizes the resulting error distributions for GPT-5-mini and Finance-Qwen. GPT-5-mini produces correct solutions for most cases, with remaining failures concentrated in numerical computation and conceptual reasoning.

Other error types are rare, and no hallucinated or unparseable outputs are observed on this subset. In contrast, Finance-Qwen exhibits errors on most evaluated cases, with failures spread across multiple categories. Conceptual and computational errors dominate, while a substantial fraction involves formatting issues or hallucinated information. Compared to GPT-5-mini, Finance-Qwen errors are less concentrated in a single mode and reflect more diverse failure behaviors. Examples of representative errors are provided in Appendix F.2, and the expanded six-model comparison is reported in Appendix F.3. Overall, this analysis shows that symbolic financial reasoning errors are heterogeneous and model-dependent, with weaker models showing more diverse and compounding failure modes, which explains the aggregate performance gaps.

6 Conclusion and Future Work

We introduced FINCHAIN, a symbolic benchmark for verifiable Chain-of-Thought financial reasoning, spanning 58 topics across 12 domains and three difficulty levels. To support step-level evaluation, we proposed CHAINEVAL, which jointly assesses intermediate reasoning consistency and final-answer correctness. Our results showed that while frontier LLMs perform best overall, even the strongest models struggle with complex symbolic reasoning, and fine-tuned open-source systems narrow down, but do not close this gap.

In future work, we plan to extend FINCHAIN to multilingual and region-specific settings and to incorporate problems grounded in real-world financial documents. This direction aims to bridge symbolic reasoning and factual verification (Xie et al., 2025), advancing more interpretable and reliable financial AI systems.

Limitations

This work has several limitations, which can be addressed in future research. First, our dataset is entirely synthetic and generated from symbolic templates. While this design enables controllable, contamination-free generation and automatic verification of both the reasoning chain and the final answer, it may lack the linguistic diversity and contextual richness of real-world financial texts. Future work could incorporate real financial documents as seed inputs for semi-structured generation while preserving symbolic grounding.

Second, the benchmark focuses on symbolic numerical reasoning and does not capture qualitative or strategic aspects of financial decision-making (e.g., risk assessment or market sentiment). Thus, FINCHAIN does not cover the full spectrum of financial reasoning, and extending it to higher-level reasoning remains an open challenge.

Third, FINCHAIN is limited to English and U.S.-centric financial conventions, limiting its applicability to multilingual and regional contexts. Extending it to other languages and financial systems is an important direction for future work.

Finally, our evaluation relies on parsing model-generated reasoning chains, which is sensitive to formatting variations and extraneous text. Improving step-level alignment via more structured outputs or tighter integration with symbolic execution is a promising direction for future work.

Ethical Statement and Broad Impact

We use only synthetic data generated via templated code and language models, without any private, sensitive, or copyrighted content. The benchmark promotes transparency and reproducibility in financial AI, but caution is needed when deploying LLMs in real-world financial decision-making, especially where correctness and regulatory compliance are critical. We hope FINCHAIN supports research toward more interpretable, verifiable, and safe reasoning systems in high-stakes domains.

Data License The FINCHAIN dataset and accompanying code is released under the MIT License.

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A FINCHAIN Construction Details

A.1 Template Creation Prompt

To construct symbolic financial reasoning benchmarks analogous to GSM-Symbolic, we design a structured prompt that guides the generation of executable financial templates. These templates support variable-based instantiation, symbolic step-wise supervision, and controlled perturbations for robustness evaluation. Below, we present the prompt used for template construction.

System Instruction: You are a financial NLP expert developing symbolic reasoning datasets. Your task is to construct symbolic templates for financial reasoning problems. Each template should support (i) controlled generation of diverse question instances, (ii) executable reasoning traces for automatic verification, and (iii) systematic variation in surface and logical complexity.

Please follow the steps below:

1. **Identify a financial reasoning task:** For example, compound interest, IRR, ROI, NPV, breakeven analysis, loan amortization, etc.
2. **Write a natural language question template:** Formulate the question using variable placeholders instead of fixed values. For instance, use {principal}, {rate}, {years}, etc.
3. **Define variables and constraints:** Specify the domain (e.g., numerical range or categorical values) for each variable. Add algebraic constraints to ensure the question is solvable and the answer valid.
4. **Write a symbolic solution trace:** Provide a step-by-step solution using the variables. Ensure the reasoning chain is executable in Python for automatic evaluation.
5. **Vary difficulty levels:** For each task, generate 10 templates with increasing complexity. Longer and more compositional reasoning chains should correspond to harder levels.

A.2 Template Examples

Here, we present example templates of three compound interest (CI) financial questions, grouped by difficulty level, including basic, intermediate, and advanced.

Basic Level

```
def template_ci_quarterly_compounding():
    """Basic: Compound Interest with Quarterly
    Compounding"""
    investor_name = random.choice(
        investor_names)
    project_name = random.choice(
        project_names)

    # Parameters
    principal = random.randint(1_000, 7_000)
    # $
    rate = round(random.uniform(2, 8), 2)
    # annual %, two decimals
    time = random.randint(1, 3)
    # years
    n = 4
    # quarterly

    # ----- Question -----
    question = (
        f"{investor_name} invests ${principal}
        in {project_name}. "
        f"The account earns {rate:.2f}%
        interest per year, compounded quarterly, "
        f"for {time} years. What is the total
        compound interest earned "
    )

    # ----- Reasoning -----
    # Step 1: future (compound) amount
    future_value = principal * (1 + rate /
        (100 * n)) ** (n * time)
    # Step 2: compound interest
    ci = future_value - principal

    # Round only for display
    fv_display = f"${future_value:.2f}"
    ci_display = f"${ci:.2f}"

    # ----- Solution -----
    solution = (
        "Step 1. Compute the future value with
        quarterly compounding:\n"
        "    n = 4 periods per year.\n"
        "    Future Value =  $P \times (1 + r / (100 \times n))^{(n \times t)}$ \n"
        f"        = ${principal} \times (1 +
        {rate:.2f}% / (100 \times 4))^{(4 \times {time})}\n"
        f"        = ${principal} \times (1 +
        {rate / (100 * n):.4f})^{4*{time}}\n"
        f"        = {fv_display}\n\n"
        "Step 2. Find the compound interest
        earned:\n"
        "    Compound Interest = Future Value -
        Principal\n"
        f"        = {fv_display} -
        ${principal}\n"
        f"        = {ci_display}"
    )

    return question, solution
```

Intermediate Level

```
def template_ci_rate_and_total_known():
    """Intermediate: Compound Interest with
    nominal rate, time, and frequency known"""

    investor_name = random.choice(
        investor_names)
    project_name = random.choice(
        project_names)

    # ----- Parameters -----
    total_amount = random.randint(5_000, 15_000)
    # Final amount A ($)
    rate = round(random.uniform(2, 10), 2)
    # Nominal annual rate %
    time = random.randint(1, 5)
    # Years
    freq_name, n = random.choice(
        [("semi-annually", 2), ("quarterly", 4), ("monthly", 12)])

    # ----- Question -----
    question = (
        f"{investor_name} received a total amount of ${total_amount:,.2f} "
        f"from their investment in {project_name}. "
        f"The investment grew at a nominal annual interest rate of {rate:.2f}% "
        f"compounded {freq_name} for {time} years. "
        f"Calculate the compound interest earned (in dollars).")

    # ----- Reasoning -----
    # Step 1: periodic rate and growth factor
    periodic_rate = round(rate / 100 / n, 6)
    # r_p
    growth_factor = round((1 + periodic_rate) ** (n * time), 6)
    # Step 2: principal P
    principal = round(total_amount / growth_factor, 2)
    # 2 dp dollars
    # Step 3: compound interest CI
    ci = round(total_amount - principal, 2)
    # ----- Solution -----
    solution = (
        "Step 1 – Find the periodic rate and growth factor\n"
        f"Periodic rate = {rate:.2f}% ÷ {n} = {periodic_rate*100:.4f}%\n"
        f"Growth factor = (1 + {periodic_rate:.6f})^{n*time} = {growth_factor:.6f}\n\n"
        "Step 2 – Compute the initial principal\n"
        f"P = A ÷ growth factor = "
        f"${total_amount:,.2f} ÷ {growth_factor:.6f} = ${principal:,.2f}\n\n"
        "Step 3 – Calculate the compound interest\n"
        f"CI = A – P = ${total_amount:,.2f} – ${principal:,.2f} = ${ci:,.2f}"
    )

    return question, solution
```

Advanced Level

```
def template_ci_with_additional_deposit():
    """Advanced: Compound Interest with a Mid Term Additional Deposit (needs 4 steps)"""
    investor_name = random.choice(
        investor_names)
    project_name = random.choice(
        project_names)

    # --- parameters ---
    principal = random.randint(2000, 8000)
    # initial $
    rate = round(random.uniform(3, 10), 2)
    # % p.a.
    time = random.randint(3, 7)
    # total years (>=3 so a mid deposit makes sense)
    n = random.choice([1, 2, 4, 12])
    # compounds per year

    deposit = random.randint(500, 4000)
    # extra $
    deposit_time = random.randint(1, time - 1)
    # year when deposit is made

    # ----- Question -----
    question = (
        f"{investor_name} initially invested $ {principal} in {project_name} at an annual "
        f"rate of {rate:.2f}%, compounded {n} times a year, for a total of {time} years. "
        f"Exactly {deposit_time} years after the start, they added an extra ${deposit} "
        f"to the same account under the same rate and compounding frequency. "
        f"Calculate the total compound interest earned by the end of the {time} years.")

    # ----- Reasoning -----
    # Step 1 – periodic rate
    periodic_rate = round(rate / (100 * n), 4)

    # Step 2 – grow the original principal for the full period
    periods_principal = n * time
    fv_principal = round(principal * (1 + periodic_rate) ** periods_principal, 2)

    # Step 3 – grow the later deposit for the remaining (time - deposit_time) years
    remaining_years = time - deposit_time
    periods_deposit = n * remaining_years
    fv_deposit = round(deposit * (1 + periodic_rate) ** periods_deposit, 2)

    # Step 4 – combine amounts and find compound interest
    total_future_value = round(fv_principal + fv_deposit, 2)
    total_contributions = principal + deposit
    compound_interest = round(
        total_future_value - total_contributions, 2)
```

```
# ----- Solution -----
solution = ("Step 1 – Periodic rate:\n"
    f" r = {rate:.2f}% / (100 x {n}) = {
    periodic_rate:.4f}\n\n"
    "Step 2 – Future value of the original
    principal:\n"
    f" Periods = {n} x {time} = {
    periods_principal}\n"
    f" FV1 = ${principal} x (1 + {
    periodic_rate:.4f})^{periods_principal} =
    "
    f"${fv_principal:.2f}\n\n"
    "Step 3 – Future value of the
    additional deposit:\n"
    f" Remaining years = {time} - {
    deposit_time} = {remaining_years}\n"
    f" Periods = {n} x {remaining_years}
    = {periods_deposit}\n"
    f" FV2 = ${deposit} x (1 + {
    periodic_rate:.4f})^{periods_deposit} = "
    f"${fv_deposit:.2f}\n\n"
    "Step 4 – Total compound interest:\n"
    f" Total FV = FV1 + FV2 = ${
    fv_principal:.2f} + ${fv_deposit:.2f} = "
    f"${total_future_value:.2f}\n"
    f" Contributions = ${principal} + ${
    deposit} = ${total_contributions}\n"
    f" Compound Interest = Total FV -
    Contributions = "
    f"${total_future_value:.2f} - ${
    total_contributions} = ${compound_interest
    :.2f}")

return question, solution
```

A.3 Data Validation Criteria

Directly prompting large language models to generate symbolic financial reasoning templates can lead to inconsistencies or incomplete specifications. To address these issues, we apply the following validation constraints prior to expert review.

Cross-national inconsistencies. Generated questions occasionally contained country-specific financial conventions (e.g., currencies, indices, or terminology). All such cases are standardized to U.S.-based financial settings.

Precision mismatch. In some cases, displayed values were rounded while computations used full precision. We align computational outputs with the displayed numerical precision.

Incomplete input specification. Some questions omitted variables required for computation. These cases are revised to include all necessary inputs.

Unit consistency. Currency symbols and units were inconsistently applied across questions and solutions. All templates are standardized to consistent units.

Non-informative steps. Certain generated solutions decomposed simple calculations into trivial substeps or omitted intermediate reasoning. These solutions are revised to reflect substantive reasoning steps.

Multiple targets. Some templates requested multiple output values, complicating evaluation. We constrain all templates to require a single target.

A.4 Expert Review and Annotation Protocol

To further enhance data quality, we recruited ten financial experts to review all validated templates. The expert panel consists of seven graduate students in economics, finance, and related quantitative disciplines, and three industry professionals with experience in quantitative research, financial engineering, and risk management. Annotators were selected through an internal vetting process to ensure domain expertise and professional credibility. Demographic details are provided in § B.1.

Annotation Platform. We developed a Streamlit-based annotation platform to facilitate efficient expert review. Implementation details are provided in § B.2.

Pilot Study. Prior to full annotation, we conduct a pilot study in which 20 templates are reviewed by all annotators to calibrate evaluation standards. After calibration, all annotators agreed on the correctness of the pilot templates.

Main Annotation. The remaining 270 templates are randomly distributed among annotators, with each template reviewed by a single expert. Out of 290 total templates, 29 are identified as incorrect and subsequently revised by financial experts. Summary statistics of identified issues are reported in § B.3.

B Annotation and Quality Control

B.1 Financial Expert Demography

To ensure the reliability and domain robustness of our benchmark, all annotations were conducted by a diverse team of financial experts and advanced students with strong quantitative and economic backgrounds. The annotators collectively represent three major categories: (1) industry professionals in quantitative research and financial engineering, (2) postgraduate students specializing in finance, economics, and auditing, and (3) experienced annotators trained in data labeling and financial analysis.

Several annotators have extensive industry experience across financial technology, quantitative research, and trading, with prior roles in investment banks, hedge funds, and fintech companies. Others are graduate students conducting research in finance, economics, and auditing, contributing academic rigor and theoretical grounding. Together, they bring complementary expertise that enhances both the practical and analytical aspects of our benchmark construction.

Summary. Our benchmark construction relies on a team of ten highly qualified annotators, including three industry professionals with prior experience in quantitative research or trading, and seven academic annotators who are graduate students in finance, economics, and auditing. This balanced composition, encompassing strong and diverse backgrounds in computer science, mathematics, statistics, and finance, ensures both professional authenticity and academic depth. Their combined expertise provides a robust foundation for high-quality, domain-consistent annotations, contributing to the overall reliability of FINCHAIN. The following are the details for each of them.

Annotator A: Currently pursuing a Ph.D. at a leading university in Asia, this annotator previously worked as a quantitative researcher at a fintech company, with experience across multiple financial markets including domestic equities, U.S. equities, Hong Kong equities, and cryptocurrencies. Their research focused on financial data generation, risk modeling, and trading strategies. They have also served as a research lead in risk management at a cryptocurrency investment fund. This blend of academic research and cross-market industry practice enhances the robustness and domain relevance of the benchmark annotations.

Annotator B: A Master’s student at a leading university with a strong undergraduate background in finance. They previously interned in the equity financing division of a major securities firm, contributing practical insights into capital markets and investment banking.

Annotator C: A Master’s student at a top institution, holding a bachelor’s degree in economics. Their training bridges theoretical economics and applied policy research, enriching the annotation process with domain-specific understanding.

Annotator D: Holds a bachelor’s degree in economics and has received graduate admission offers from top international institutions.

Their interdisciplinary background strengthens the dataset’s coverage of trade and international finance contexts.

Annotator E: Holds a bachelor’s degree in economics, providing a solid foundation in macroeconomic theory and financial principles that supports reliable annotation and consistency across financial texts.

Annotator F: A Master’s student at a well-known university specializing in auditing and intelligent systems, with a research focus on large language model evaluation and its applications in auditing. Their familiarity with both auditing and financial concepts supports the annotation of financial news and auditing benchmarks from a research-oriented perspective.

Annotator G: A Master’s student at a university recognized for its auditing and financial programs, with strong grounding in auditing, financial analysis, and data quality control. Their prior participation in annotation projects ensures consistent standards for annotation accuracy.

Annotator H: A quantitative analyst with an MSc-equivalent degree in financial technology from a top UK university. They have prior experience at major global financial institutions, focusing on stochastic modeling, risk management, and process automation. They also contribute to research on large language models in finance and are advancing toward professional certification in investment analysis.

Annotator I: A quantitative researcher at a global investment firm with prior experience at quantitative research and technology companies. Their work spans cross-asset systematic strategies, portfolio optimization, and machine learning applications in trading. They also serve as a teaching assistant for a postgraduate course on systematic trading strategies.

Annotator J: A quantitative trading analyst focused on equity derivatives, holding a postgraduate degree in financial engineering and risk management from a top European university and a bachelor’s degree from a globally recognized institution. Their professional experience includes roles at several financial institutions across asset management, banking, and fintech, covering alpha-signal development, portfolio optimization, and derivatives trading.

B.2 Annotation Platform

We developed a custom annotation platform to evaluate the correctness of Python templates that generate financial questions and solutions. Each template corresponds to a financial scenario (e.g., investment analysis, compound interest, deposits, or ratio calculations). Annotators are instructed to review the code and determine whether both the financial framework and its implementation are correct, and whether the output representation (e.g., units, rounding) complies with the annotation policy.

The annotation task requires a binary verdict: *Correct* or *Incorrect*. Templates labeled as *Correct* need no modifications, though annotators may optionally provide comments. Templates labeled as *Incorrect* must be associated with one or more issue tags, accompanied by a minimal code correction and a brief explanation. A template is considered *Correct* when its financial framework, calculations, and representation fully conform to the policy. It is marked as *Incorrect* if any substantive flaw is present in framework selection, mathematical logic, representation, robustness, or clarity. To facilitate consistent labeling, we introduced five issue tags:

- **Formula Choice Error:** An incorrect financial framework or formula is applied (e.g., simple vs. compound interest).
- **Math/Logic Error:** Arithmetic or algorithmic errors within the chosen formula (e.g., $r \times n$ instead of r/n).
- **Representation Error:** Inconsistent or incorrect handling of numbers, units, or rounding (e.g., annual vs. monthly mismatch).
- **Robustness Error:** Failures on boundary or extreme inputs (e.g., division by zero, negative values).
- **Clarity Issue:** Ambiguous variable names or comments that hinder auditability, even if the numerical results are correct.

To further support annotators, the platform provides curated reference cases across five templates within a single finance topic: compound interest. Each case includes (1) a question example, (2) a potential error type aligned with one of the defined issue tags, (3) a bad solution illustrating the error, and (4) a minimal code fix.

Issue Type	Count	Proportion (%)
Representation Error	12	41.4
Clarity Issue	9	31.0
Formula Choice Error	5	17.2
Math/Logic Error	3	10.3
Robustness Error	2	6.9
Total Tagged Templates	29	100.0

Table 3: Distribution of issue types among annotated templates.

Figure 6 shows two representative cases: a *Formula Choice Error*, where simple interest is incorrectly applied in a compound interest setting, and a *Math/Logic Error*, where the exponent is omitted. Such examples provide concrete guidance for annotators, ensuring consistency and reliability.

After reviewing these reference cases, annotators proceed to the main annotation interface, where they evaluate unseen templates (Figure 7). For each template, annotators must issue a binary verdict, select one or more issue tags if applicable, and provide a minimal code correction with a short justification. The interface presents the Python template and its generated question on the left, while the right panel allows annotators to record their verdict, choose tags, and edit the code directly. This design mirrors realistic auditing conditions and ensures that annotations capture both error identification and corrective reasoning.

B.3 Annotated Template Issue Statistics

Out of 290 templates, 29 (10%) were tagged as containing errors during the annotation process. We summarize the distribution of issue types among annotated templates in Table 3.

Most problems stem from representation and clarity errors, followed by formula selection, logical inconsistencies, and robustness issues.

B.4 Review Rubrics

To ensure fair and interpretable human evaluation, each model response is assessed along two complementary dimensions: *Reasoning Process Quality* and *Final Answer Accuracy*. For each question, reviewers are provided with the question itself, the standard reference answer, and the generated responses from different models. They independently assign scores on a 1–5 scale for each dimension, following the detailed rubrics below.

Simple CI Calculation (annual compounding): Template Questions & Tagged Error Cases

Question example

{Investor} invested \$P in Project X. The investment grows at an annual interest rate of r% compounded annually over t years. Calculate the compound interest.

Cases

- Uses simple interest inside a CI template

Tag: Formula Choice Error

Why: Selected simple-interest framework $(1+r \cdot t)$ instead of compounding $(1+r)^t$.

Bad solution:

$$A = P \cdot (1 + r/100 \cdot t)$$

$$CI = A - P$$

Minimal fix:

$$A = P \cdot (1 + r/100)^t$$

$$CI = A - P$$

- Exponent omitted

Tag: Math/Logic Error

Why: Forgot to raise to the power t.

Bad solution:

$$A = P \cdot (1 + r/100)$$

$$CI = A - P$$

Minimal fix:

$$A = P \cdot (1 + r/100)^t$$

Figure 6: Reference examples for compound interest templates, illustrating typical annotation cases with error tags, flawed solutions, and minimal fixes.

B.5 Reasoning Process Quality

This dimension evaluates how clearly, logically, and correctly the model articulates its reasoning steps leading to the final answer. High-quality reasoning should demonstrate coherent logical flow, factual correctness, and consistency with valid domain principles.

- 1 (Unacceptable):** Illogical, incoherent, or irrelevant reasoning; missing steps or severe conceptual errors.
- 2 (Poor):** Some reasoning attempt but with major factual or procedural flaws; inconsistent or unclear Chain-of-Thought.
- 3 (Fair):** Partial understanding with mixed correct and incorrect reasoning; superficial or incomplete explanation.
- 4 (Good):** Mostly correct and coherent reasoning with minor inaccuracies or unclear phrasing; logical flow generally sound.
- 5 (Excellent):** Clear, well-structured, and logically consistent reasoning throughout; fully correct and well-justified steps.

B.6 Final Answer Accuracy

This dimension evaluates the correctness and completeness of the model’s final output relative to the reference solution. Reviewers compare each model’s final answer with the standard answer to determine whether the model’s conclusion is correct and sufficiently supported.

- 1 (Unacceptable):** Completely incorrect or missing answer; no alignment with the reference solution.
- 2 (Poor):** Largely incorrect due to major conceptual or computational errors.
- 3 (Fair):** Partially correct; captures some relevant elements but omits or distorts key aspects of the correct solution.
- 4 (Good):** Largely correct and complete with only minor inaccuracies that do not affect the main result.
- 5 (Excellent):** Fully correct, precise, and complete; matches the reference solution exactly or with an equivalent formulation.

FinChain — Expert Verification

☐ Show guidelines

Progress

0/290

Jump to Review ID

Review ID

◀ Prev

Jump

Next ▶

Review ID: 1

```
def template_security_intermediate_reg_a_investment_limit():
    """3:Intermediate: Check if investment exceeds Reg A+ Tier 2 limit (4 reasoning steps)"""

    investor, company = random_entities()
    annual_income = random.randint(50_000, 150_000)
    investment = random.randint(20_000, 70_000)
    is_audited = random.choice([True, False])
    limit_ratio = 0.10
    limit = round(annual_income * limit_ratio, 2)

    question = (
        f"{investor} wants to invest ${investment:,} in a Regulation A+ Tier 2 offering from {company}.\n"
        f"Their reported annual income is ${annual_income:,}. "
        f"The offering is {'audited' if is_audited else 'unaudited'}.\n"
        f"Under SEC rules, unaudited offerings are subject to a 10% income cap. "
        f"Can {investor} legally make this investment?"
    )

    # Determine if limit applies and assess legality
    solution = (
        f"Step 1: Identify audit status → Offering is {'audited' if is_audited else 'unaudited'}.\n"
        f"Step 2: Determine if 10% income cap applies → "
        f"{'No cap for audited offerings.' if is_audited else f'Cap applies → 10% of ${annual_income:,}'}\n"
        f"Step 3: {'No comparison needed (✓)' if is_audited else f'Compare investment ${investment:,} to limit ${limit:,}'}\n"
        f"Step 4: Conclusion → "
        (
            "Yes, the investment is allowed because the offering is audited."
            if is_audited else (
                "Yes, investment is within the permitted cap."
                if investment <= limit else
                "No, investment exceeds the allowable limit for unaudited offerings."
            )
        )
    )

    return question, solution
```

Review & Edit

Verdict

☐ Correct
☒ Incorrect

Issue tag definitions

Select one or more issue tags (required):

Choose an option

Edited code (required)

```
1 def template_security_intermediate_reg_a_investment_limit():
2     """3:Intermediate: Check if investment exceeds Reg A+ Tier 2
3     limit (4 reasoning steps)"""
4
5     investor, company = random_entities()
6     annual_income = random.randint(50_000, 150_000)
7     investment = random.randint(20_000, 70_000)
8     is_audited = random.choice([True, False])
9     limit_ratio = 0.10
10    limit = round(annual_income * limit_ratio, 2)
11
12    question = (
13        f"{investor} wants to invest ${investment:,} in a
14        Regulation A+ Tier 2 offering from {company}.\n"
15        f"Their reported annual income is ${annual_income:,}. "
16        f"The offering is {'audited' if is_audited else
17        'unaudited'}.\n"
18        f"Under SEC rules, unaudited offerings are subject to a
19        10% income cap. "
20        f"Can {investor} legally make this investment?"
21    )
22
23    # Determine if limit applies and assess legality
24    solution = (
25        f"Step 1: Identify audit status → Offering is {'audited'
26        if is_audited else 'unaudited'}.\n"
27        f"Step 2: Determine if 10% income cap applies → "
28        f"{'No cap for audited offerings.' if is_audited else
29        f'Cap applies → 10% of ${annual_income:,} = ${limit
```

Figure 7: Expert annotation interface. Annotators review each template, assign a verdict, select issue tags, and provide minimal code corrections.

C Model Detail Information

Table 4 provides details about the evaluated models.

D Metric Evaluation and Ablations

In order to better understand the behavior of our proposed evaluation measure, we conducted a series of ablation experiments and comparative analysis. All quantitative results reported in this section are benchmarked against expert human evaluations of reasoning quality (see § B.1 for expert details), ensuring robustness and consistency across settings.

D.1 Ablations of the DTW-Based Metric

Our main evaluation metric, the **Normalized DTW Alignment Score (Gate Mode)**, measures both local semantic-numeric agreement and global sequence-level alignment between predicted and gold reasoning traces.

To assess its robustness and the effect of its design choices, we considered several variants:

- **DTW Gate Mode.** This is the primary formulation used in the paper. Semantic similarity and numeric agreement are combined multiplicatively, i.e., $\text{Score}_{\text{gate}}(i, j) = \text{SS}(s_i^*, \hat{s}_j) \times \text{AM}(s_i^*, \hat{s}_j)$. This “gating” ensures that steps contribute only when semantic meaning and intermediate results align consistently across diverse reasoning scenarios.
- **DTW Soft Mode.** A more permissive variant that blends semantic and numeric agreement through a weighted combination: $\text{Score}_{\text{soft}}(i, j) = \alpha \text{SS}(s_i^*, \hat{s}_j) + \beta \text{AM}(s_i^*, \hat{s}_j)$, with $\alpha = 0.85$ and $\beta = 0.15$. This “soft” formulation captures cases where partial numeric agreement still reflects correct reasoning, providing smoother sensitivity to small deviations. In other words, while the Gated version will assign 0 to a sequence of aligning reasoning steps, which resulted in a wrong answer (which can be a case if an LLM fails mathematics behind the solution), Soft version will still give a higher score.

Model	Organization	Size	Backbone	Source
Frontier Proprietary LLMs				
GPT-5	OpenAI	N/A	–	gpt-5-2025-08-07
GPT-4.1	OpenAI	N/A	–	gpt-4.1-2025-04-14
GPT-5-mini	OpenAI	N/A	–	gpt-5-mini-2025-08-07
GPT-4.1-mini	OpenAI	N/A	–	gpt-4.1-mini-2025-04-14
Claude Sonnet 4.5	Anthropic	N/A	–	claude-sonnet-4-5-20250929
Claude Sonnet 4	Anthropic	N/A	–	claude-sonnet-4-20250514
Claude Sonnet 3.7	Anthropic	N/A	–	claude-3-7-sonnet-20250219
Gemini-2.5 Pro	Google	N/A	–	Last Update: June 2025
Gemini-2.5 Flash	Google	N/A	–	Last Update: June 2025
DeepSeek-V3.2	DeepSeek	N/A	–	Last Update: Sep 29 2025
DeepSeek-V3.1	DeepSeek	N/A	–	Last Update: Sep 22 2025
DeepSeek-R1	DeepSeek	N/A	–	Last Update: Jan 20 2025
Grok-4 Heavy	xAI	N/A	–	grok-4-0709
Grok-4 Fast	xAI	N/A	–	grok-4-fast-reasoning
Finance Specific LLMs				
Fin-o1	TheFinAI	8B	meta-llama/Llama-3.1-8B	TheFinAI/Fin-o1-8B
Fin-R1	SUFE-AIFLM-Lab	7B	Qwen/Qwen2.5-7B-Instruct	SUFE-AIFLM-Lab/Fin-R1
DianJin-R1	Qwen DianJin Team	7B	Qwen/Qwen2.5-7B-Instruct	DianJin/DianJin-R1-7B
Finance-LLaMA	Wiro AI	8B	deepseek-ai/DeepSeek-R1-Distill-Llama-8B	WiroAI/WiroAI-Finance-Llama-8B
Finance-Qwen	Wiro AI	7B	Qwen/Qwen2.5-7B	WiroAI/WiroAI-Finance-Qwen-7B
Math Enhanced LLMs				
WizardMath	WizardLM Team	7B	mistralai/Mistral-7B-v0.1	WizardLMTeam/WizardMath-7B-V1.1
MetaMath	MetaMath Project	7B	EleutherAI/l1emba 7b	meta-math/MetaMath-7B-V1.0
Mathstral	Mistral AI	7B	mistralai/Mistral-7B-v0.1	mistralai/Mathstral-7B-v0.1
Qwen-2.5-Math	Qwen Team	7B	Qwen/Qwen2.5-7B	Qwen/Qwen2.5-Math-7B-Instruct
General Purpose Open LLMs				
LLaMA-3.1	Meta	8B	–	meta-llama/Llama-3.1-8B
Qwen-2.5	Qwen Team	7B	–	Qwen/Qwen2.5-7B-Instruct
Qwen-3	Qwen Team	8B	–	Qwen/Qwen3-8B

Table 4: Details of the organization and model source (i.e. model version for proprietary models, and HuggingFace model name for open-source models) for the LLMs evaluated in FINCHAIN.

- **DTW Precision, Recall, and F1.** In addition to the normalized alignment score, we derive DTW-based precision, recall, and F1 measures that quantify step-level coverage under the DTW alignment path. These provide a finer breakdown of reasoning alignment.

To capture final correctness, we tried a weighted sum of DTW-based scores and final answer correctness. We varied α in the range 0.1–0.9 to identify the best proportion. Based on these experiments, $\alpha = 0.1$ yielded the best Spearman ρ and was used for comparison.

D.2 Comparative Evaluation

We also evaluated a range of traditional text-similarity and reasoning metrics, including ROUGE-2, ROUGE-L, step-level precision and recall (marked as ‘non-DTW’ in the table), BERTScore, and our weighted sum of DTWNormGate and final answer’s correctness (DTWNormGate+FAC in the table). Each metric was correlated with expert-assigned *Reasoning Process Quality* score. Table 5 summarizes the top Spearman correlations with expert process judgments.

Metric	Spearman ρ
DTWNormGate+FAC	0.655
DTWNormGate	0.640
DTW Precision (Soft)	0.625
DTW Precision (Gate)	0.622
DTW F1 (Soft)	0.619
DTW F1 (Gate)	0.618
Step Precision (non-DTW)	0.604
DTWNormSoft	0.592
DTW Recall (Gate)	0.573
Step Recall (non-DTW)	0.570
DTW Avg. Path Score (Gate)	0.529
DTW Avg. Path Score (Soft)	0.526
DTW Recall (Soft)	0.512
ROUGE-2	0.469
ROUGE-L	0.434
BERTScore	0.287

Table 5: Spearman correlation with expert evaluation of the reasoning process.

As additional validation, we also measure other correlation metrics: Kendall tau (which measures similarity of rankings) and Pearson correlation. As shown in Table 6 and Table 7, our suggested metric still holds high correlation position even with other correlation metrics across different evaluation settings and conditions.

Metric	Pearson ρ
DTWNormGate+FAC	0.584
DTW F1 (Soft)	0.584
DTW Precision (Soft)	0.578
DTW F1 (Gate)	0.568
DTW Avg. Path Score (Gate)	0.556
DTW Recall (Gate)	0.550
Step Recall (non-DTW)	0.549
DTW Precision (Gate)	0.548
Step Precision (non-DTW)	0.530
DTWNormSoft	0.515
DTW Avg. Path Score (Soft)	0.515
DTW Recall (Soft)	0.511
DTWNormGate	0.492
ROUGE-L	0.421
ROUGE-2	0.420
BERTScore	0.300

Table 6: Pearson correlation with expert evaluation of the reasoning process.

Metric	Kendall τ
DTW Precision (Gate)	0.511
DTWNormGate+FAC	0.509
Step Precision (non-DTW)	0.505
DTW F1 (Gate)	0.503
DTWNormGate	0.503
DTW Precision (Soft)	0.494
DTW F1 (Soft)	0.487
Step Recall (non-DTW)	0.477
DTW Recall (Gate)	0.463
DTWNormSoft	0.460
DTW Avg. Path Score (Gate)	0.420
DTW Avg. Path Score (Soft)	0.401
DTW Recall (Soft)	0.389
ROUGE-2	0.360
ROUGE-L	0.330
BERTScore	0.213

Table 7: Kendall τ correlation with expert evaluation of the reasoning process.

D.3 Sensitivity to the Numerical Tolerance ϵ

To assess the robustness of ChainEval to the numerical tolerance parameter ϵ , we vary ϵ over a broad range and measure the Spearman correlation between the resulting metric scores and expert human judgments on reasoning quality. The results are shown in Table 8.

The results show that the correlation remains relatively stable across a broad range of tolerance values. Performance peaks at $\epsilon = 0.05$, which is also consistent with commonly used materiality thresholds in financial auditing. Overall, the variation is modest, suggesting that the main conclusions are not sensitive to a specific threshold choice.

ϵ	Spearman ρ
0.00	0.54
0.01	0.60
0.03	0.60
0.05	0.64
0.15	0.61
0.20	0.61
0.50	0.60
0.75	0.59

Table 8: Sensitivity analysis of the numerical tolerance parameter ϵ in ChainEval. We report the Spearman correlation between metric scores and expert human judgments.

D.4 Discussion

The DTW-based variants consistently achieve the highest correlation with expert judgments, with the **Normalized DTW Alignment Score (Gate Mode) with FAC** emerging as the most reliable indicator of reasoning faithfulness. Non-FAC and the ‘‘Soft’’ variant yields slightly lower but still strong correlations, suggesting that the gating formulation better captures strict consistency, while the soft variant provides smoother sensitivity to near-correct reasoning. Compared to traditional metrics such as ROUGE or simple step-level precision and recall, DTW captures not only semantic similarity but also the structural coherence and numerical consistency of reasoning chains. These results highlight the usefulness of our proposed metric.

D.5 Suggestions For Reproducibility

Although our approach shows a strong correlation with human evaluations, it is not without limitations. The way in which the reasoning steps are parsed from the model’s output plays an important role in the quality estimation. In this work we combine both LLM-based parsing and parsing using regular expressions. We try to split the response on reasoning steps using regular expressions, which capture ‘Step X’-like patterns in the response, if such patterns are not found, we instruct LLM to split the answer on steps, copy-pasting the whole step string. We also use LLM to parse the final answer from the response. We acknowledge that such approaches depend on the prompting strategy, and there is a chance that other parsing and comparison methods (like the use of executable symbolic engine such as SymPy) can produce varying results. For better reproducibility we share our parsing model details and prompts used to extract final answer and reasoning steps.

We used the GPT-4.1 model. Results were parsed using OpenAI’s output schema, which allowed us to avoid inconsistency in outputs. For the system prompt we used the following text:

You are a strict financial reasoning parser. Your task is to convert a noisy, long, conversational Chain-of-Thought solution into a clean and structured JSON object. Follow these rules exactly:

1. No Calculations
Do NOT compute anything.
Do NOT recompute numbers from the text.
Do NOT round, simplify, or adjust any numeric value.
2. Number Extraction (CRITICAL)
You must copy numbers EXACTLY as they appear in the text.
Keep the sign intact (do NOT remove or change negative signs).
Keep decimal precision exactly as written.
Remove commas, dollar signs, and percent signs only as formatting, not as value changes.
3. Step Extraction
Identify only the actual reasoning steps.
Ignore filler text: summaries, meta-comments, confidence statements, restatements, chatter.
Produce concise step descriptions capturing the logic of each step.
For each reasoning step, output an object with:
{
 "index": <step_number>,
 "text": "<short description of the step>",&br/> "value": <numeric result of that step or null>
}
4. How to Determine value
If the step contains a computation result, copy the final numeric result in that step.
Usually this is the last number in the step (e.g., number after "=").
If the step has no numeric result, set "value": null.
5. Final Answer Extraction
"final_answer" must be copied exactly as the last numeric value in the entire solution.
Apply the same numeric-copy rules as above.
6. JSON Output Format
Return ONLY a valid JSON object of the form:
{
 "steps": [...],
 "final_answer": <number>
}
7. Forbidden Behaviors
Do NOT change "-0.2636" into "2636.0".
Do NOT remove negative signs.
Do NOT round 1.514016 to 1.51.
Do NOT compute new numbers.
Do NOT invent numbers not present in the text.
Do NOT choose earlier numbers if a later number is clearly the result.
Accuracy is measured strictly.
Copy numbers exactly.

Model	CHAIÑEVAL	FAC	ROUGE R ₂
Grok 4 Heavy	65.64	81.00	23.87
GPT-5	68.42	69.50	28.37
Gemini 2.5 Pro	67.92	73.00	18.85
Fin-R1	56.44	34.00	5.43
Mathstral	64.02	51.00	18.99

Table 9: **Performance comparison on a randomly sampled 200-instance subset using CHAIÑEVAL, final answer correctness (FAC), and ROUGE R₂.** These results are reported for reference only and are not directly comparable to full-benchmark results.

For the user prompt we used:

Here is the full solution text to parse into reasoning steps and a final answer:
f"{text}\n\n"
Remember: follow the instructions and return ONLY the JSON object.

In case the steps were parsed with regular expression correctly, the final answer was retrieved using the following prompt:

You are given a full solution text. Ignore step segmentation. Your ONLY job is to extract 'final_answer' as the last numeric value in the solution, following the same numeric-copy rules from the system prompt. Return ONLY a JSON object of the form:
"{ \"final_answer\": <number or null> }"
Here is the solution text:

E Complementary Results

E.1 Results on a Sampled Subset

Due to the high inference cost of Grok 4 Heavy, we evaluate this model on a randomly sampled subset of 200 instances rather than the full benchmark. The results in Table 9 are reported to provide a coarse reference for this cost-limited setting and should not be directly compared with the full-benchmark results in Table 2.

E.2 Complementary Domain-Level Results

We report additional domain-level results for the remaining models grouped by model category. These figures are provided for completeness and transparency, and no further analysis is conducted. Figure 8 reports domain-level CHAIÑEVAL scores for additional frontier proprietary models, showing broadly consistent performance patterns across financial domains with moderate variation. Figure 9 presents domain-level performance for finance-tuned models, illustrating heterogeneous patterns across domains.

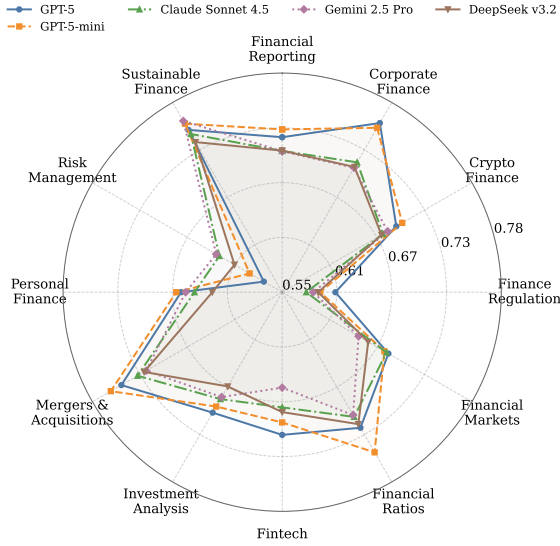


Figure 8: **Domain-level performance of frontier proprietary models.** Radar plot showing CHAINEVAL scores across twelve financial domains for GPT-5, GPT-5-mini, Claude Sonnet 4.5, Gemini 2.5 Pro, and DeepSeek v3.2.

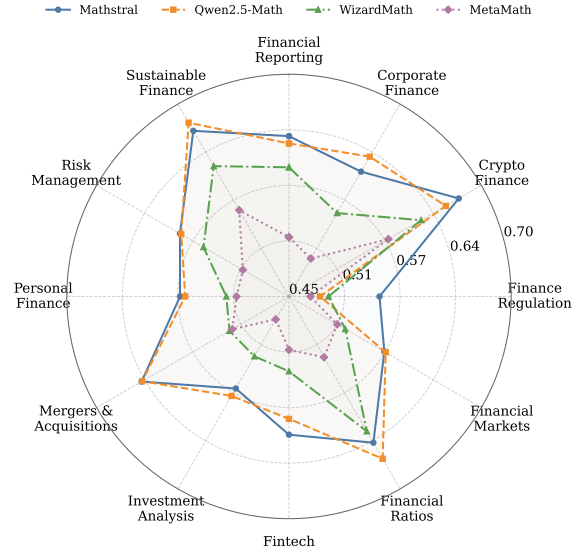


Figure 10: **Domain-level performance of math-enhanced models.** Radar plot showing CHAINEVAL scores across twelve financial domains for Mathstral, Qwen2.5-Math, WizardMath, and MetaMath.

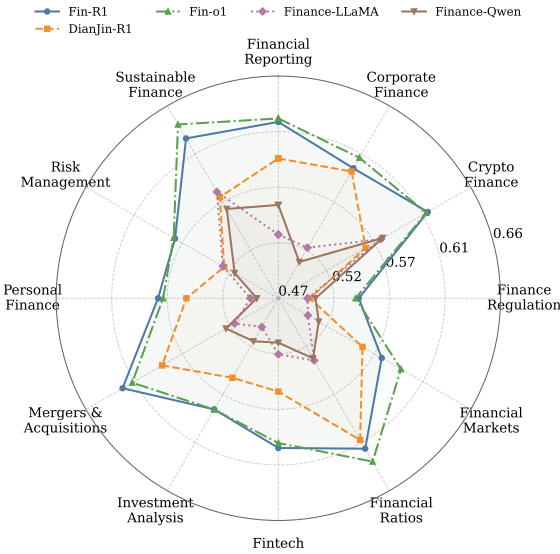


Figure 9: **Domain-level performance of finance-tuned models.** Radar plot showing CHAINEVAL scores across twelve financial domains for Fin-R1, DianJin-R1, Fin-o1, Finance-LLaMA, and Finance-Qwen.

Figure 10 shows domain-level results for math-enhanced models, with performance varying across financial domains. Figure 11 reports domain-level performance for general-purpose open models, serving as reference baselines across domains.

E.3 Difficulty Breakdown under CHAINEVAL

As shown in Figure 12, CHAINEVAL exhibits a more gradual change across difficulty tiers than FAC. This indicates that models may maintain partial step-level alignment on harder instances even when end-task success decreases, consistent with CHAINEVAL assigning partial credit to intermediate reasoning.

F Error Analysis Supplementary

F.1 Error Taxonomy

This appendix defines the error categories used in the diagnostic error analysis. Each model output is assigned to a single category based on the primary criterion violated in the predicted reasoning or final result.

No Error. The reasoning process and final result match the gold computation within the predefined tolerance.

Computational. The applied formula or method is appropriate, but one or more intermediate or final numerical computations are incorrect.

Conceptual. The reasoning violates a problem constraint or applies a financial rule or formula inconsistently with the gold specification.

Sign / Direction. The sign or directional definition of a quantity is inconsistent with the gold formulation, such as reversing subtraction order or treating a decrease as an increase.

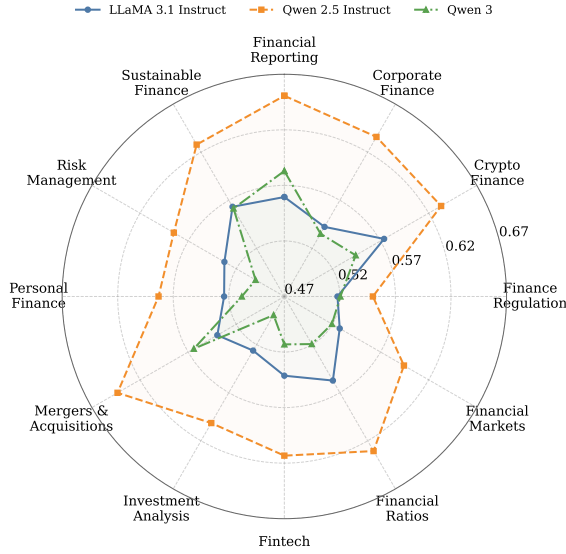


Figure 11: **Domain-level performance of general-purpose open models.** Radar plot showing CHAINEVAL scores across twelve financial domains for LLaMA 3.1 Instruct, Qwen 2.5 Instruct, and Qwen 3.

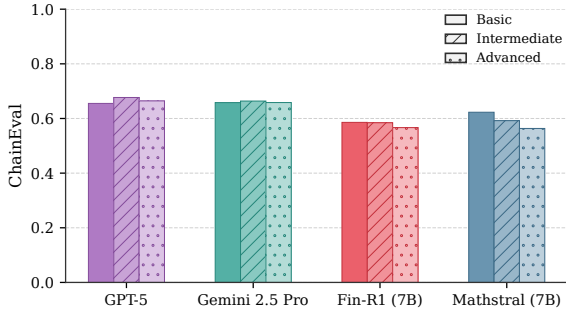


Figure 12: **CHAINEVAL across difficulty levels.** CHAINEVAL scores for representative models on *Basic*, *Intermediate*, and *Advanced* FINCHAIN instances. Compared to FAC in the main text, CHAINEVAL varies more gradually across difficulty, reflecting partial credit from step-level alignment.

Unit / Precision. Units, scale, or numerical precision are handled inconsistently with the problem specification, for example percent versus decimal or dollars versus millions.

Parsing / Format. The output structure is malformed or inconsistent, preventing reliable parsing or alignment of reasoning steps.

Hallucination. One or more variables, assumptions, or quantities are introduced that are not supported by the original question or provided context.

F.2 Error Examples

Table 10 presents one representative example for each error category used in the diagnostic analysis. Each example consists of a short excerpt from the model output and a concise description of the specific criterion under which the output is labeled as erroneous. The descriptions focus on observable mismatches between the model output and the corresponding gold computation or definition, such as numerical inconsistency, violation of problem constraints, or unsupported quantities. The examples are intended solely to illustrate how the error taxonomy is applied in practice.

They do not aim to explain the underlying causes of model behavior, assess error frequency, or attribute failures to model training, architecture, or reasoning capacity.

F.3 Expanded Error Analysis

To test whether the qualitative findings from the main-text error analysis generalize beyond two representative systems, we extend the same annotation protocol to six models spanning different performance levels: GPT-5-mini, Finance-Qwen, DeepSeek v3.2, Fin-R1, Mathstral, and Qwen 2.5 Instruct. All models were evaluated on the same randomly sampled subset of 200 questions, ensuring a controlled and directly comparable evaluation setting across systems.

The full results are presented in Table 11, which provides a detailed breakdown of error categories for each model.

The expanded comparison yields the same qualitative pattern as in the main text. Stronger models tend to fail primarily on computational and conceptual mistakes, typically arising from arithmetic inaccuracies or incorrect application of financial formulas, whereas lower-performing models exhibit a broader and more heterogeneous error profile, including substantially more parsing/format issues and hallucinated reasoning steps that are unsupported by the input.

Overall, the expanded analysis strengthens the robustness of our qualitative findings and shows that the observed error patterns are not limited to two representative models, but generalize consistently across models with varying performance levels and training paradigms.

Error Type	Model	Model Output (snippet)	Error Description
Computational	GPT-5-mini	“Total cash = $3,592,610.3 \times 1.8632 = \$6,693,751.51$.”	Incorrect multiplication; correct total is approximately \$6.68M.
Conceptual	GPT-5-mini	“ $47408 + x = 0.8(60427 + x) \Rightarrow x = 4668$.”	Violates conservation of total portfolio value during rebalancing.
Sign / Direction	GPT-5-mini	“Change = new - original ≈ -0.123 .”	Uses an inconsistent sign convention; gold definition is old - new.
Unit / Precision	GPT-5-mini	“175 (whole credits) ... total rebate = $175 \times 2.03 = \$355.25$.”	Applies unjustified rounding to a fractional quantity (175.5 credits).
Parsing / Format	Finance-Qwen	“... on the investment at the end of 3 years?”	Output is malformed and cannot be reliably parsed into steps.
Hallucination	Finance-Qwen	“Assume 500M shares ... Market cap = $\text{EPS} \times \text{P/E} = \33.74B .”	Introduces an unsupported quantity and applies an inconsistent valuation formula.

Table 10: Representative error examples with model outputs (expert-audited sample).

Error Type	GPT-5-mini	Finance-Qwen	DeepSeek v3.2	Fin-R1	Mathstral	Qwen 2.5 Instr.
No Error	80.5% (161)	10.0% (20)	70.5% (141)	39.0% (78)	23.5% (47)	34.0% (68)
Computational	7.5% (15)	24.5% (49)	18.0% (36)	35.5% (71)	36.0% (72)	44.0% (88)
Conceptual	9.5% (19)	31.0% (62)	8.5% (17)	18.0% (36)	30.5% (61)	17.0% (34)
Sign/Direction	1.0% (2)	0.0% (0)	0.5% (1)	0.5% (1)	2.0% (4)	0.5% (1)
Unit/Precision	1.5% (3)	2.0% (4)	1.5% (3)	4.5% (9)	6.0% (12)	2.0% (4)
Parsing/Format	0.0% (0)	15.5% (31)	1.0% (2)	1.0% (2)	1.5% (3)	1.0% (2)
Hallucination	0.0% (0)	17.0% (34)	0.0% (0)	1.5% (3)	0.5% (1)	1.5% (3)
Error Rate	19.5% (39)	90.0% (180)	29.5% (59)	61.0% (122)	76.5% (153)	66.0% (132)

Table 11: Expanded error analysis across six models on the same sampled set of 200 questions.